

MODEL QUESTION PAPER

REVISION-2010
SUB CODE:3065

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

BASIC MEDICAL SCIENCE
(Common for BM AND MD)

Time 3hrs

Maximum Marks : 100

PART - A

Answer all Questions in one or two sentences. Each question carries **2 marks**.

I.

1. Define Biochemistry and Biophysics?
2. What is cardiac output?
3. Define residual volume?
4. What do you mean by pH of a solution?
5. List the electrodes used for ECG measurement?

5x2=10

PART - B

Answer any **FIVE** of the following questions. Each question carries **6 marks**.

II.

1. Explain the structure of a nerve cell with a neat sketch?
2. Describe the temperature regulation of human body?
3. Explain the basic concepts of hemodialysis?
4. Describe the digestion of proteins and lipids?
5. Explain the morphology of ECG wave?
6. Explain the mechanism of equilibrium maintenance in human body?
7. Describe the various sensory receptors?

5x6=30

PART - C

Answer one full question from each Unit. Each question carries 15 marks.

UNIT I

- III** a) Explain elementary tissues of the body? **10**
b) Describe monophasic and biphasic rehrscording? **5**

OR

- IV.** Explain the origin of Biopotentials with neat diagrams? **15**

UNIT II

- V.** Explain the lead configurations used in electrocardiography with necessary diagrams? **15**

OR

- VI** a). Explain the central nervous system with the structure of Brain? **12**
b) Write the application of heat and cold in medicine? **3**

UNIT III

- VII** a). Draw the structure of nephron and explain? **12**
b) Draw the spirogram and mark the different parameters? **3**

OR

- VIII** a) Explain the mechanism of hearing with a neat sketch? **10**
b) Explain the action of rods and cones in the mechanism of vision? **5**

UNIT IV

- IX** . Explain the process of metabolism of carbohydrates, fats and proteins? **15**

OR

- X** a) Draw and explain human digestive system? **10**
b) Explain the pH measurement technique? **5**

15x4=60